Emre Can Kadirhan

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SKILLS

Programming Languages: Python, Java, JavaScript, TypeScript, C, C++
Frameworks: Django, FastAPI, React, Spring Boot, Next.js
Tools and Technical Skills: Git/GitHub, Docker, CI/CD, AWS, Linux/Unix, Postman, PyTest, Junit, PostgreSQL, MySQL, Prisma, SQLAlchemy, Redis, Celery

PROJECTS

Bot Battle Arena | Python, FastAPI, Celery, Redis, PostgreSQL, Docker | **GitHub**

- Engineered a referee system that orchestrates Docker containers, executing untrusted bot code in fully isolated sandboxes with disabled networking, memory caps, PID limits, and dropped Linux capabilities to prevent abuse
- Built an async match execution pipeline using Celery and Redis, fully decoupling match orchestration from the API layer to prevent container workloads blocking web requests
- Implemented real-time match streaming using FastAPI WebSockets and Redis pub/sub, bridging the Celery worker and API server processes to broadcast live game state to connected viewers
- Designed a PostgreSQL schema with turn-by-turn state snapshots enabling complete match replay
- Implemented an ELO rating algorithm to rank bots after each match based on rating difference and outcome

SwingVision - AI Golf Swing Analyzer | Next.js, Python, PostgreSQL, Redis, MediaPipe, BetterAuth, Prisma| GitHub

- Built a Python worker using OpenCV and MediaPipe to overlay real-time AI pose skeletons on user uploaded golf swing videos to analyze swing mechanics
- Integrated Redis based async task queue to offload video processing, keeping web response times under 200ms during heavy load
- Architected a PostgreSQL (Neon) schema to manage user accounts, video metadata, and professional coaching feedback with relational integrity
- Implemented role-based access control using Next.js and BetterAuth, supporting separate user and admin dashboard experiences

EXPERIENCE

TQueue Software StartUp **Istanbul, Turkey**
Software Engineer Intern *June 2024 – August 2024*

- Developed and maintained RESTful APIs using Spring Boot for a customer queue management system that served as the primary API layer for core product
- Refactored the customer management module into a more service-oriented architecture, reducing technical debt
- Wrote unit and integration tests using JUnit and Mockito, increasing test coverage on the customer service module from near zero to roughly 70%

EDUCATION

Arizona State University – Tempe, AZ B.S. in Computer Science | 2025 - 2027 | GPA: 3.86
The University of Texas at Dallas – Richardson, TX B.S in Computer Science | 2022 – 2025